

FIGURE 1. Step 0 (Early training). Side-by-side histograms of windowed word counts. **Left:** Incorrect words show sub-Poisson clustering with narrow distributions, reflecting repetitive short fragments. **Right:** Correct words are extremely sparse and scattered, displaying near-Poisson randomness.

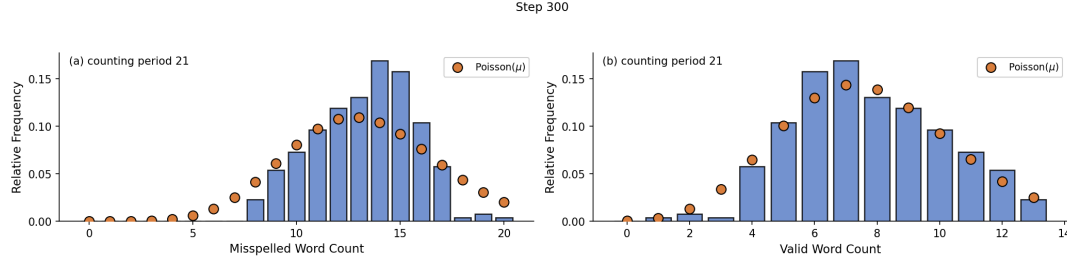


FIGURE 2. Step 300 (Mid-training, approaching transition). **Left:** Incorrect words maintain sub-Poisson clustering with narrow, variance-suppressed distributions as repetitive fragments persist. **Right:** Correct words begin showing nascent structure but remain scattered and near-Poisson.

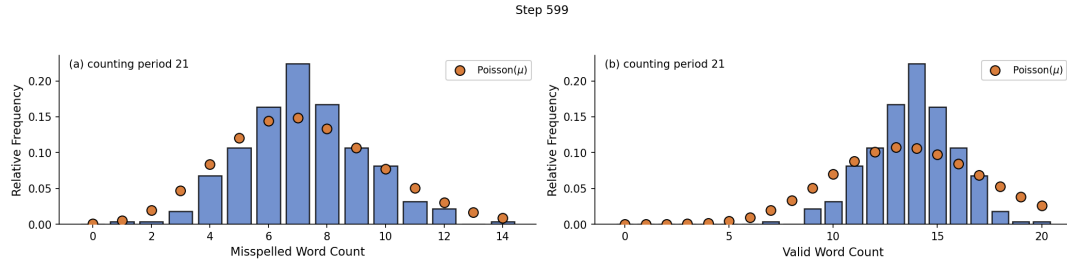


FIGURE 3. Step 599 (Post-transition). The dispersion flip is complete. **Left:** Incorrect words have broadened to near-Poisson distributions, becoming sparse, independent residual errors that occur randomly. **Right:** Correct words have contracted into tight, sub-Poisson distributions, reflecting regular, structured usage.

E. KL DIVERGENCE FROM POISSON

While dispersion captures deviations through the variance-to-mean ratio, it cannot detect all distributional changes. To complement this analysis, we compute the Kullback–Leibler (KL) divergence between the empirical histogram $\hat{p}(n)$ of windowed counts and a fitted Poisson distribution with mean $\hat{\lambda}$:

$$D_{\text{KL}}(\hat{p} \parallel \text{Pois}(\hat{\lambda})) = \sum_n \hat{p}(n) \log \frac{\hat{p}(n)}{P(n; \hat{\lambda})}. \quad (4)$$

Here $D_{\text{KL}} = 0$ only if the empirical distribution matches the Poisson baseline exactly. KL divergence measures the full distributional deviation from Poisson, providing sensitivity to higher-order statistical structure beyond variance alone. Together, dispersion and KL divergence form complementary probes: dispersion directly quantifies the regime (Poisson vs. sub-Poisson), while KL divergence quantifies the magnitude of deviation. The synchronization of cusps in both metrics strengthens evidence for genuine reorganization rather than metric artifacts.

F. ESTIMATION AND VISUALIZATION

All metrics are averaged across 5 seeds, with shaded error bands (where plotted) denoting $\pm$ one standard deviation. Histograms are shown at representative early and late checkpoints. All decoding parameters are fixed unless otherwise noted.

IV. RESULTS

A. OVERVIEW OF DISCONTINUITIES

Across all five seeds, every external metric exhibits a coordinated, transition-like reorganization that is not visible in the smooth training/validation losses. In the same epoch band, we observe: (i) synchronized cusps in both dispersion and KL divergence; (ii) a temporary reversion of correct-word dispersion toward Poisson before stabilizing to sub-Poisson; (iii) a rise–then–collapse of the unique incorrect vocabulary alongside steady growth of the unique correct vocabulary; and (iv) an S-curve jump in average correct-word length from ~ 1.5 to ~ 2.5 characters. Qualitatively, the model transitions from producing short fragments and errors

to composing multi-character, coherent words. The convergence of multiple independent, continuous metrics—not just one—supports the interpretation of a genuine phase-transition-like reorganization rather than a gradual drift or metric artifact.

B. WORD-FREQUENCY SNAPSHOTS AT 150, 250, 350, AND 500

To visualize the lexical reorganization concretely, we examine frequency distributions of the top generated words at four representative checkpoints spanning the transition (Figs. 4–7). These snapshots reveal the progression from fragmentary, error-dominated output to coherent, structured vocabulary.

Step 150 (Pre-transition): Generated text is dominated by short fragments and malformed tokens. The most frequent outputs are single characters or defective two-letter combinations that do not correspond to valid English words. Error tokens (shown in red) outnumber correct words, and even the correct tokens tend to be single-character artifacts like punctuation or isolated letters. The model has not yet discovered multi-character compositional structure.

Step 250 (Transition onset): Early coherent words begin to appear alongside persistent errors. Legitimate two- and three-letter words such as “to”, “is”, and “my” emerge with increasing frequency, but the distribution remains mixed. Fragmentary errors still account for a substantial fraction of outputs, indicating the model is in an exploratory phase where valid patterns coexist with residual noise. The balance between correct and incorrect vocabulary is shifting but not yet stable.

Step 350 (Transition band): Mid-length words such as “you”, “the”, and “and” emerge decisively and stabilize at the top of the frequency distribution. Error tokens collapse in both diversity and frequency, now appearing primarily as low-frequency outliers. The shift from single-character fragments to multi-character words is nearly complete. This checkpoint corresponds precisely to the epochs where dispersion and KL divergence exhibit synchronized cusps, confirming that the statistical reorganization manifests as a lexical restructuring.

Step 500 (Post-transition): Coherent words dominate the distribution, and their relative frequencies have settled into stable patterns reflecting the underlying corpus statistics. High-frequency function words (“the”, “and”, “you”) appear consistently, while error tokens are sparse and relegated to the tail. The vocabulary has consolidated around legitimate forms, and word lengths reflect compositional structure rather than random character sequences.

Together, these snapshots provide concrete lexical evidence for the exploratory → consolidating shift captured by our Poisson-based metrics. The visual progression from fragment-dominated to word-dominated output directly parallels the dispersion flip (incorrect sub-Poisson → Poisson, correct near-Poisson → sub-Poisson) and the KL divergence

cusps, demonstrating that the statistical reorganization has clear linguistic correlates.

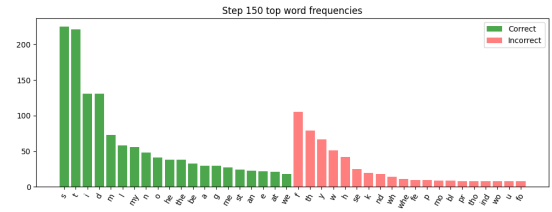


FIGURE 4. Step 150 (Pre-transition). Top generated words are dominated by short fragments and errors (red bars). Single characters and malformed tokens account for most outputs. The model has not yet learned multi-character compositional structure.

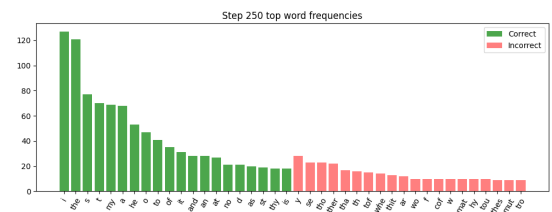


FIGURE 5. Step 250 (Transition onset). Early coherent words (“to”, “is”, “my”) appear alongside persistent errors. The model is in an exploratory regime where valid patterns coexist with fragmentary noise.

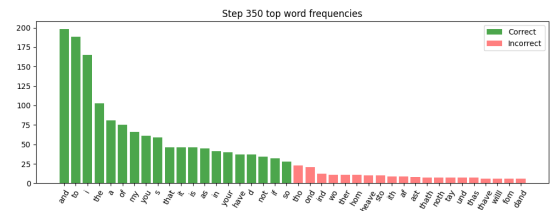


FIGURE 6. Step 350 (Transition band). Mid-length words (“you”, “the”, “and”) emerge decisively. Error diversity collapses. This checkpoint aligns with the synchronized cusps in dispersion and KL divergence.

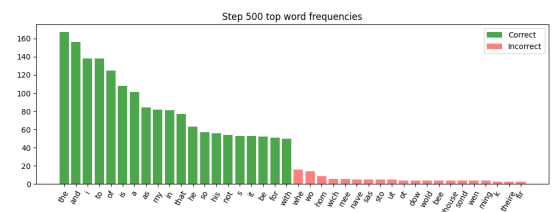


FIGURE 7. Step 500 (Post-transition). Coherent words dominate with stable frequency patterns. High-frequency function words are consistent; errors are sparse. The vocabulary has consolidated around legitimate forms.

C. AVERAGE WORD LENGTH (CORRECT WORDS)

Average correct-word length provides a direct linguistic indicator (Fig. 8). After a long, flat phase near single-character outputs (~ 1.5), the mean jumps in an S-curve to ~ 2.5 characters in the same transition band highlighted

later by dispersion and KL divergence. The sharp increase signals the onset of composing multi-character words and is consistent with the subsequent sub-Poisson regularity of correct usage.

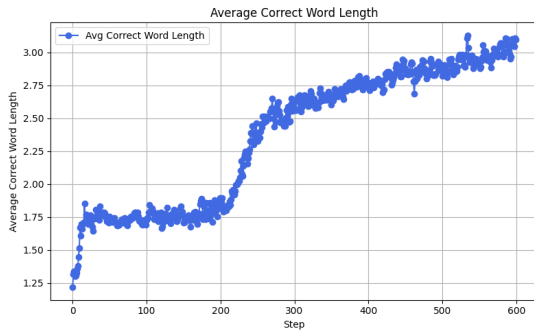


FIGURE 8. Average correct-word length vs. epoch. The S-curve jump aligns with the transition band identified by both dispersion and KL divergence.

D. UNIQUE VOCABULARY DYNAMICS

Unique-type dynamics separate exploration from consolidation. The incorrect vocabulary (Fig. 9) exhibits a plateau (steps 0–200), a transient peak near step 250 (fragment proliferation), followed by a sustained collapse as erroneous forms are abandoned. This peak coincides precisely with the transition band identified by dispersion and KL divergence.

The correct vocabulary (Fig. 10) displays multi-phase growth: rapid initial accumulation (steps 0–100), deceleration and plateau (steps 100–250), sharp acceleration during the transition (steps 250–400), and continued growth post-transition. The acceleration phase aligns with the dispersion and KL cusps, indicating that the statistical reorganization manifests as expansion of the legitimate vocabulary.

The push–pull dynamics—incorrect vocabulary peaks as correct vocabulary accelerates—suggest that fragment proliferation precedes word consolidation: the model explores erroneous combinations before committing to valid forms.

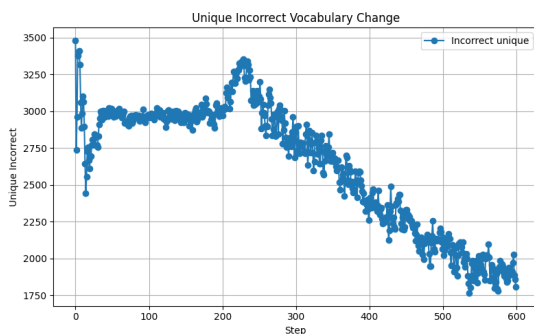


FIGURE 9. Unique incorrect vocabulary: plateau → peak near step 250 → collapse. The peak coincides with the transition band.

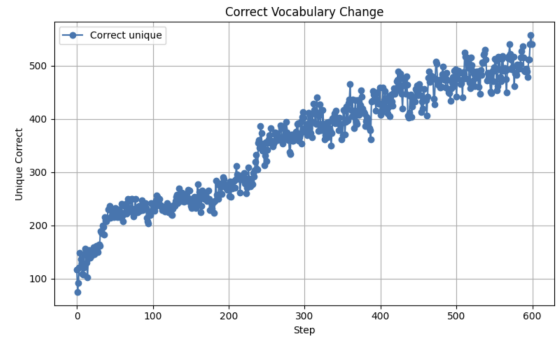


FIGURE 10. Unique correct vocabulary: multi-phase growth with acceleration during the transition (steps 250–400) aligning with dispersion and KL divergence cusps.

E. CASE STUDY: PREFIX FORMATION AND ACQUISITION OF “YOU”

To test whether the transition corresponds to composing longer words from shorter fragments, we track the evolution of the word “you” and its prefixes. In Fig. 11, counts of *y*, *yo*, and *you* are plotted across steps. Early training is dominated by single-letter *y*; as the transition approaches, *yo* appears transiently; at the transition, the full word *you* rises sharply while *y* and *yo* decline. This pattern is consistent with consolidation of fragments into a stable three-letter form. Fig. 11 also shows the learning curve for *you* itself: an S-shaped rise precisely in the band where dispersion and KL divergence display cusps. In Poisson terms, the temporary reversion of correct-word dispersion toward $D \approx 1$ can be interpreted as a brief exploratory phase during recombination; post-transition, correct usage settles into sub-Poisson regularity while residual errors are Poisson-like.

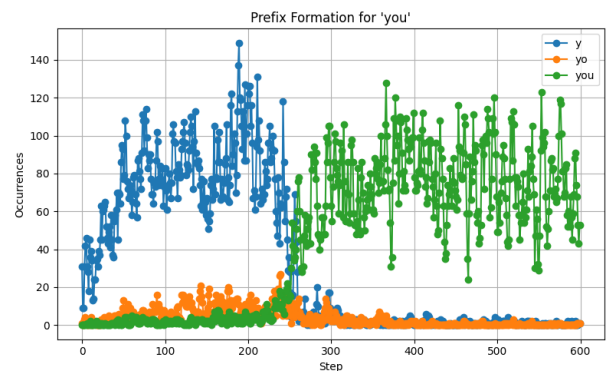


FIGURE 11. Prefix formation for “you”: counts of *y*, *yo*, and *you* across training steps. Near the transition, the full word *you* rises sharply while prefix counts decline, indicating consolidation from fragments into a multi-character word.

F. KL DIVERGENCE TO POISSON BASELINE

KL divergence provides an independent measure of how much the empirical word-count distributions deviate from the Poisson baseline. We track correct and incorrect words separately to reveal the divergence–convergence duality that characterizes the transition.